

Topic 9: Introduction to Kinetics and Equilibria

9A — Kinetics

9.1 — Collision Theory and Factors Affecting Rate

A. Collision Theory: The Basis of Reaction Rates

For a chemical reaction to occur, **particles must collide** with:

1. **Sufficient energy** to overcome the **activation energy (E_a)**
2. **Correct orientation** to break and form new bonds

Key term — Activation Energy (E_a):

The **minimum energy** that particles must have in order for a successful reaction to occur.

If particles collide:

- **With $E \geq E_a$** → reaction occurs (successful collision)
- **With $E < E_a$** → particles bounce off (no reaction)

B. Factors That Affect the Rate of a Chemical Reaction

Each factor increases rate by **increasing frequency of collisions, energy of collisions, or percentage of successful collisions.**

Factor	Effect on Particles	Reason for Increased Rate
Concentration (of solutions)	More particles per unit volume	More frequent collisions
Pressure (of gases)	Same number of particles in smaller volume	Higher collision frequency
Temperature	Particles move faster and have more energy	More collisions with $E \geq E_a$
Surface area (solids)	More particles exposed for collisions	Increases frequency of collisions
Catalyst	Provides alternative pathway with lower E_a	Increases % of successful collisions

C. Key Terms

Term	Definition
Rate of reaction	Change in concentration (or amount) per unit time
Successful collision	Collision with enough energy and correct orientation
Activation energy (E_a)	Minimum energy required for particles to react
Catalyst	Substance that increases rate without being used up by lowering E_a

D. Example Scenarios

1. Dilute vs Concentrated Acid on Marble Chips

- Higher concentration = faster bubbling (faster CO_2 release)

2. Powdered vs Lump Solid

- Powder reacts faster due to increased surface area

3. Heating Hydrogen and Oxygen Mixture

- Without spark = no reaction
- With spark = explosion (overcomes E_a)

9.2 — Activation Energy and Energetic Requirements for Reactions

A. What Is Activation Energy (E_a)?

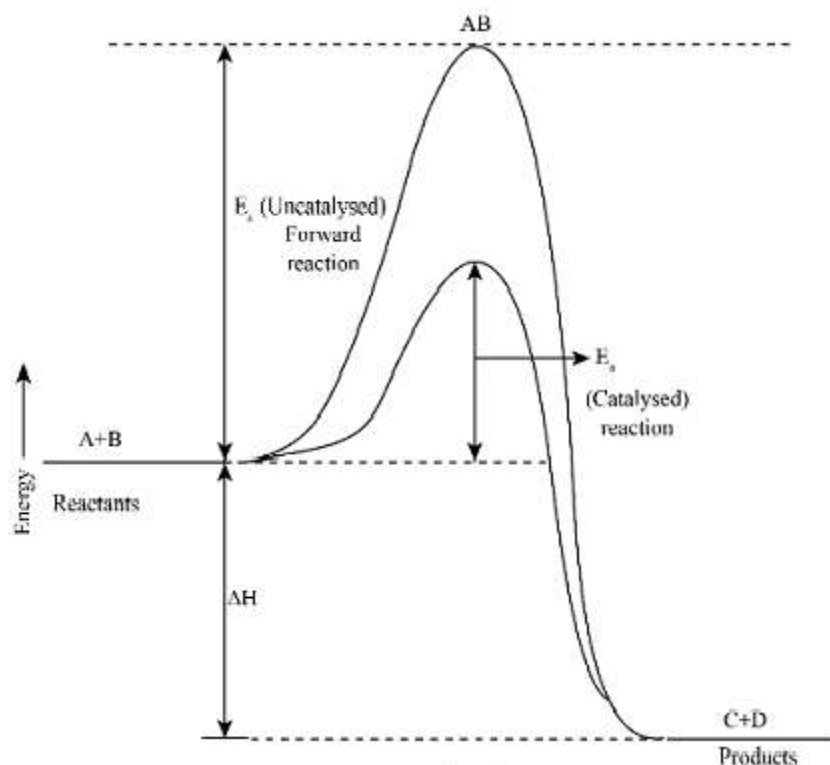
Activation energy (E_a) is the **minimum amount of energy** that colliding particles must possess for a reaction to occur.

- It is the **energy barrier** that must be overcome to break bonds in the reactants and allow new bonds to form in the products.
- Even if two particles collide, **no reaction occurs** unless their combined kinetic energy $\geq E_a$.

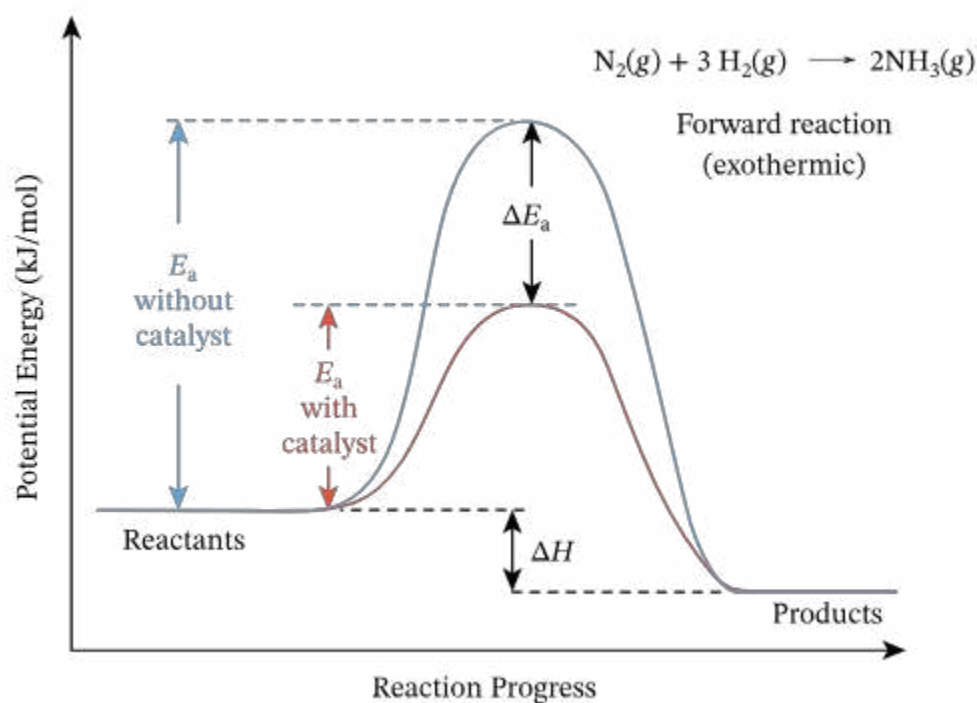
B. Visual Representation — Energy Profile

In an **energy profile diagram**, activation energy is shown as the peak between reactants and products.

For an exothermic reaction:



For an endothermic reaction:



C. Importance of Activation Energy

Scenario	Outcome
E of colliding particles $< E_a$	No reaction (particles bounce apart)
$E \geq E_a$	Reaction occurs (bonds break/form)
Lower E_a (with catalyst)	More particles have enough energy $\rightarrow$ higher rate

D. Temperature and E_a

Increasing temperature **does not lower** E_a , but:

- Increases the **kinetic energy** of particles
- More particles** now have $E \geq E_a$
- So, **rate of reaction increases**

E. Catalysts and Activation Energy

Catalysts **lower the activation energy** by providing an **alternative reaction pathway**.

- E.g., a catalyst may allow a reaction to proceed via an **intermediate**
- This means a **greater proportion** of particles have enough energy to react at a given temperature

9.3 — Measuring and Calculating Reaction Rate

A. Definition of Rate of Reaction

Rate of reaction = how quickly **reactants are used up** or **products are formed**.

B. Two Common Ways to Calculate Rate

1. Average Rate from Time Taken

Used in **simple time-based experiments**, e.g.:

- Time for gas to be collected
- Time for a cross to disappear in a precipitation reaction

Formula:

$$\text{Rate} = 1 / \text{time}$$

Units: s^{-1}

(Used when the reaction is measured by a visual endpoint)

Example:

If a reaction takes 35 s to complete:

$$\rightarrow \text{Rate} = 1 \div 35 = \mathbf{0.0286 \text{ s}^{-1}}$$

2. Rate from Graph Gradient

Used when **product/reactant quantity is recorded over time**, e.g.:

- Volume of gas vs time
- Mass lost vs time
- Concentration vs time

$$\text{Rate} = \text{gradient} = \Delta y / \Delta x$$

Units depend on what's being measured:

- Volume/time $\rightarrow \text{cm}^3 \text{ s}^{-1}$
- Concentration/time $\rightarrow \text{mol dm}^{-3} \text{ s}^{-1}$
- Mass/time $\rightarrow \text{g s}^{-1}$

C. Key Types of Graphs and Where to Measure Rate

Graph Type	What It Shows	Where to Measure Rate
Product vs Time	e.g., gas volume	Gradient = rate
Reactant vs Time	e.g., concentration	Negative gradient = rate
Curve graph	Non-constant rate	Use tangent at required point
Straight line	Constant rate	Gradient is constant

D. Initial Rate Method

Initial rate = rate at **t = 0**

Used in:

- Clock reactions
- Reactions measured via continuous monitoring

To find it:

- Draw a **tangent to the curve at t = 0**
- Calculate its gradient

E. Practical Examples

1. Measuring CO₂ gas from CaCO₃ + HCl

- Plot volume vs time
- Use gradient = $\Delta \text{volume} / \Delta \text{time} \rightarrow \text{rate in cm}^3 \text{ s}^{-1}$

2. Disappearing cross (precipitation)

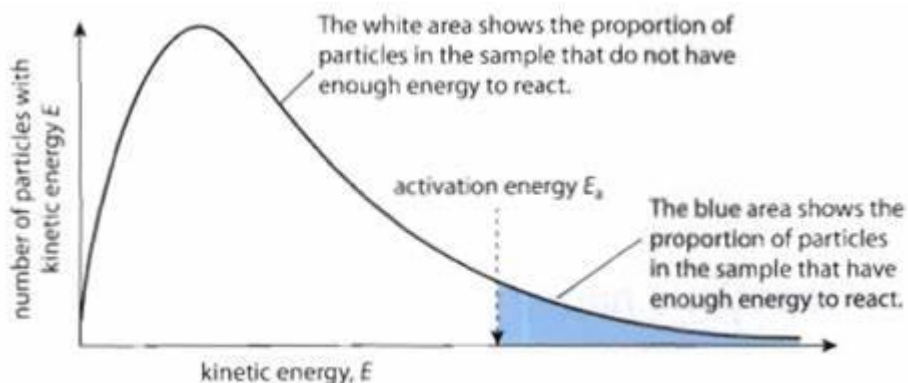
- Time taken to obscure a mark = time
- Rate = $1 / \text{time} \rightarrow \text{s}^{-1}$

9.4 — Maxwell–Boltzmann Distribution and Temperature Effect

A. What Is the Maxwell–Boltzmann Distribution?

It is a graph showing how the **kinetic energy** of particles in a sample is **distributed at a given temperature**.

B. Key Features of the Maxwell–Boltzmann Curve



Feature	Description
Starts at the origin (0,0)	No particles have zero energy
Asymmetrical peak	Most particles have moderate energy
Curve does not touch x-axis	Some particles have very high energy (but number tends to zero)
Area under the curve	Represents total number of particles (fixed for a given sample)
Area beyond E_a	Represents particles with enough energy to react

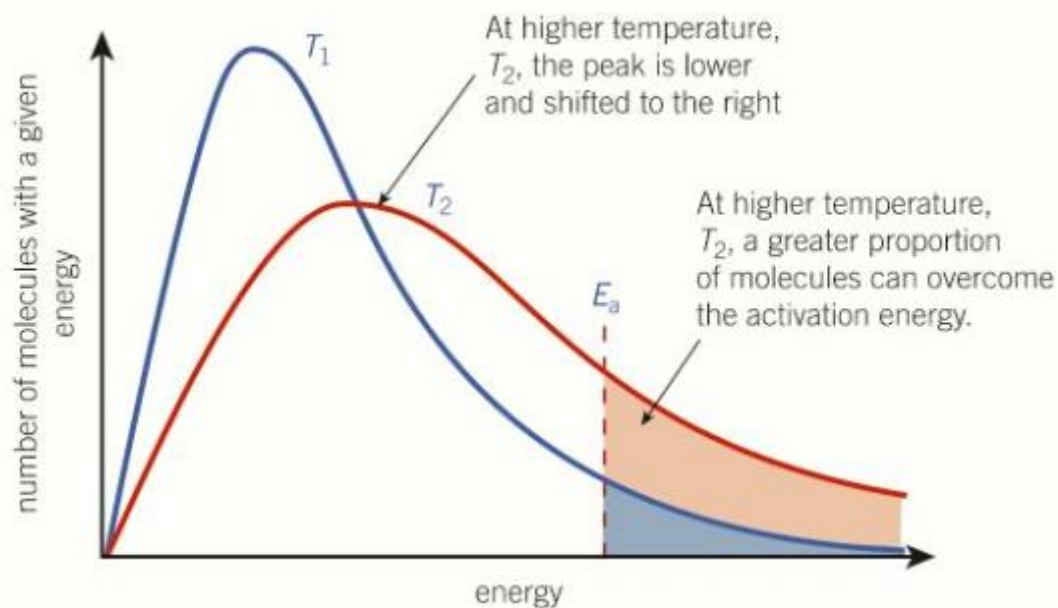
C. Effect of Temperature on the Distribution

Higher Temperature:

- Peak moves **lower and to the right**
- Curve **flattens and widens**
- Area under curve remains **constant**
- **More particles have $E \geq E_a$** → increased rate

Conclusion:

Increasing temperature significantly increases the **proportion of particles that can react**, not just their speed.



D. Comparison of Two Curves: T_1 vs T_2 ($T_2 > T_1$)

Factor	At T_1 (lower)	At T_2 (higher)
Most probable energy	Lower	Higher
Curve height	Taller	Shorter
Particles with $E \geq E_a$	Fewer	More
Rate of reaction	Slower	Faster

Important Exam Point:

Even a **small temperature increase** causes a **large increase in rate**, because of the **exponential rise** in particles with $E \geq E_a$.

9.5 & 9.8 — Catalysts and Reaction Rate

A. What Is a Catalyst?

A **catalyst** is a substance that **increases the rate of a chemical reaction** by providing an **alternative reaction pathway** with a **lower activation energy**, without being used up in the process.

- Catalysts are **not chemically changed** or consumed
- They may be recovered unchanged at the end of the reaction

B. How Catalysts Increase Reaction Rate

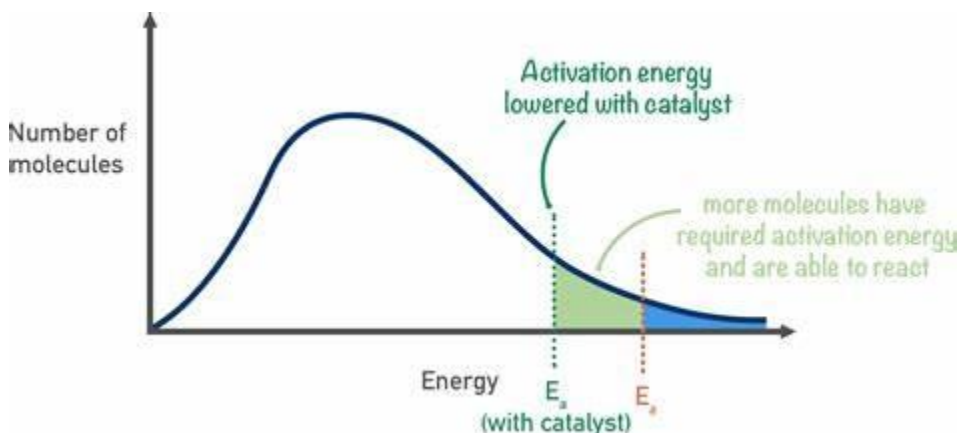
- A catalyst **lowers the activation energy (E_a)**
- This means **more particles have energy $\geq E_a$**
- Hence, the **frequency of successful collisions increases**

C. Maxwell–Boltzmann Distribution with a Catalyst

- The curve **remains unchanged**
- The E_a **line shifts leftward** (lower activation energy)
- Area under the curve to the right of E_a **increases** → more particles can react

Conclusion:

A catalyst does **not change the shape** of the curve, only the position of E_a .



D. Types of Catalysts

Type	Description	Example
Homogeneous	Catalyst is in the same phase as reactants	H ⁺ in ester hydrolysis
Heterogeneous	Catalyst is in a different phase	Solid Fe in Haber process
Biological	Enzyme (specific biological catalyst)	Amylase for starch digestion

E. Examples of Industrial Catalysts

Process	Catalyst	Purpose/Outcome
Haber process (NH ₃)	Iron (Fe)	Faster formation of ammonia
Contact process (H ₂ SO ₄)	Vanadium(V) oxide (V ₂ O ₅)	Increases SO ₃ yield
Hydrogenation of alkenes	Nickel (Ni)	Converts unsaturated fats to saturated
Catalytic converters	Platinum, rhodium	Converts CO, NO → CO ₂ , N ₂ , H ₂ O

(D and E good for knowing not necessary to remember)

9.6 — Reaction Profiles: Catalysed and Uncatalysed Pathways

Basically you have to draw or describe or explain the diagrams for the reaction profiles which are already given in 9.2B

Catalysed vs Uncatalysed Reaction Profile

When a **catalyst is added**:

- The **activation energy is lowered**
- A new **reaction pathway** appears with **intermediate steps**
- **ΔH remains the same** (because reactants and products are the same)

9.7 — Industrial Use of Catalysts and Sustainable Chemistry

A. Why Catalysts Are Important in Industry

Catalysts are used in nearly all large-scale industrial reactions because they:

1. **Increase the rate of reaction**
2. Allow reactions to occur at **lower temperatures and pressures**
3. **Reduce energy costs** and improve **efficiency**
4. Enable use of **less reactive, safer, or cheaper feedstocks**

B. Catalysts and Sustainability

Using a catalyst can improve sustainability by:

Benefit	How It Helps Sustainability
Lower temperatures needed	Reduces fossil fuel use → less CO ₂ emitted
Less waste	Improves atom economy (fewer by-products)
Reusability	Catalysts aren't used up , reducing waste
Faster reactions	Less energy consumption per unit product

Sustainability = meeting present needs without compromising future generations i.e. renewability, carbon emissions etc.

C. Examples in Industry

Process	Catalyst Used	Sustainable Impact
Haber Process (NH ₃ production)	Iron (Fe)	Lower T/P → less energy → lower CO ₂
Contact Process (H ₂ SO ₄)	Vanadium(V) oxide (V ₂ O ₅)	Increases rate → less energy needed
Hydrogenation (of alkenes)	Nickel (Ni)	Converts oils → margarine more efficiently
Catalytic Cracking (of hydrocarbons)	Zeolite	Increases yield of useful fuels
Catalytic Converters (cars)	Platinum, Rhodium	Reduces toxic emissions (CO, NO _x)

Note: you do not need to necessarily memorise the catalysts used here except for hydrogenation and catalytic cracking, these are given as examples

D. Green Chemistry Principle

Catalysts support the principles of **green chemistry** by:

- Avoiding unnecessary high temperatures/pressures
- Reducing side-products
- Enabling **selective reactions**
- Reducing **hazardous reagents** and improving safety

E. Key Exam Phrases

When asked about industrial benefits of catalysts, include points like:

- “Lower energy consumption”
- “Fewer carbon emissions”
- “Improved atom economy”
- “Reduces environmental impact”
- “Reduces costs for heating or using a high pressure or using more expensive equipment to maintain those conditions”

(9.8 is given together with 9.5 as they are related)

9B — Equilibria

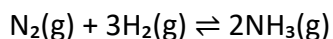
9.9 — Reversible Reactions and Dynamic Equilibrium

A. What Is a Reversible Reaction?

A reaction in which the **products can react to reform the reactants** under suitable conditions.

Symbol: $\rightleftharpoons$

Example:



B. What Is Dynamic Equilibrium?

A state in a **closed system** where:

- The **forward and reverse reactions occur at the same rate**
- The **concentrations of reactants and products remain constant**

Conditions required:

- Must be a **closed system** (no loss of substances to surroundings)
- Reaction must be **reversible**
- System must have had **enough time to reach equilibrium**

C. Key Characteristics of Dynamic Equilibrium

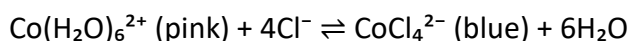
Feature	Explanation
Closed system	Prevents mass loss or gain
Microscopic activity	Forward and reverse reactions still occur
Macroscopic constancy	Concentrations of all species stay constant
Equilibrium is dynamic	Particles are still colliding and reacting

D. Misconceptions to Avoid

Incorrect Belief	Correction
"Reactions stop at equilibrium"	No — both directions continue at equal rate
"All substances are present in equal amounts"	No — concentrations are constant, not necessarily equal

E. Real-Life Example

Cobalt(II) chloride in water:



- Adding HCl shifts equilibrium to the **right** (blue)
- Adding water shifts it to the **left** (pink)

Used as a **reversible colour-change demonstration**

9.10 — Le Chatelier's Principle and Shifts in Equilibrium

A. Le Chatelier's Principle:

If a system at equilibrium is subjected to a **change in conditions**, the position of equilibrium will **shift to oppose the change**.

In other words, the system tries to **counteract any imposed stress**.

Applies only in a CLOSED system.

B. Types of Changes That Affect Equilibrium

1. **Temperature**
2. **Pressure** (gases only)
3. **Concentration**

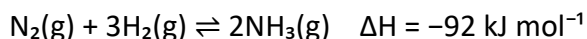
4. Catalyst (note: **does not affect position**)

C. Effect of Temperature

Depends on whether the reaction is **exothermic** ($-\Delta H$) or **endothermic** ($+\Delta H$)

Change	Exothermic Reaction ($-\Delta H$)	Endothermic Reaction ($+\Delta H$)
$\uparrow$ Temperature	Shifts left (favours reverse)	Shifts right (favours forward)
$\downarrow$ Temperature	Shifts right	Shifts left

Example:



Temperature increases $\rightarrow$ Forward reaction is exothermic $\rightarrow$ Equilibrium shifts left $\rightarrow$ **Yield of ammonia decreases**

Temperature decreases $\rightarrow$ Forward reaction is exothermic so equilibrium shift right $\rightarrow$ **Yield of ammonia increases**

D. Effect of Pressure (Only for gases)

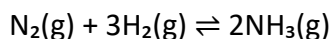
Pressure only affects reactions where the number of **moles of gas differs** on each side of the equation.

Rules:

- **If pressure is increased:**
System reduces pressure by shifting toward the side with **fewer moles of gas**
- **If pressure is decreased:**
System increases pressure by shifting toward the side with **more moles of gas**

Example:

Haber Process:



- Left side: $1 + 3 = 4$ **moles of gas**
- Right side: **2 moles of gas**
- **Increasing pressure** $\rightarrow$ shifts equilibrium **to the right** (fewer gas moles) $\rightarrow$ **more ammonia formed**
- **Decreasing pressure** $\rightarrow$ shifts equilibrium **to the left** (more gas moles) $\rightarrow$ **less ammonia formed**

Important note:

If both sides have the **same number of gas moles**, pressure change has **no effect** on equilibrium position.

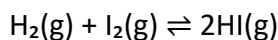
E. Effect of Concentration

Changing concentration only affects **aqueous or gaseous systems**.

General Rule:

- **Increasing the concentration of a reactant** → system shifts toward the **products** (forward direction)
- **Increasing the concentration of a product** → system shifts toward the **reactants** (reverse direction)
- **Decreasing the concentration of reactant** → shifts toward **reactants**
- **Decreasing or removing product** → shifts toward **products**

Example:



- **Add more H₂** → equilibrium shifts to the right → more HI formed
- **Remove some HI** → equilibrium shifts to the right to replace it
- **Add extra HI** → equilibrium shifts to the left → forms more H₂ and I₂

F. Effect of a Catalyst

- **No effect** on position of equilibrium
- **Speeds up** both forward and reverse reactions equally
- Helps the system **reach equilibrium faster**

Change in Condition	Direction of Equilibrium Shift	Effect on Yield
Increase in temperature (if forward is exothermic)	Shifts to left (reverse reaction favoured)	Decreases yield of product
Decrease in temperature (forward exothermic)	Shifts to right (forward reaction favoured)	Increases yield of product
Increase in pressure	Shifts to side with less moles	If the products have fewer moles , yield of product increases If the reactants have fewer moles , yield of product decreases
Decrease in pressure	Shifts to side with more moles	If the products have more moles , yield of product increases If the reactants have more moles , yield of product decreases
Add more reactant	Shifts to right	Increases product formation
Remove product	Shifts to right	Increases product formation
Add more product	Shifts to left	Decreases product concentration
Use of catalyst	No shift in equilibrium position	Product yield stays the same, but reached faster

9.11 — Industrial Compromises Between Rate and Yield

A. The Central Problem

In many industrial reversible reactions, the **conditions that favour high yield** are **not the same** as those that favour a **fast reaction rate**.

High yield may require:

- Low temperature (favourable for exothermic reactions)
- High pressure (if products have fewer gas moles)

Fast rate may require:

- High temperature
- High pressure
- Use of a catalyst

But these may:

- Be **too expensive**
- Pose **safety risks**
- **Slow down** the reaction too much at ideal yield conditions

B. The Industrial Solution: Compromise Conditions

Industries choose **conditions that balance**:

1. **Economics** (cost and profit)
2. **Safety**
3. **Sufficiently fast rate**
4. **Acceptable product yield**

C. Key Examples of Industrial Compromises

1. Haber Process (Ammonia production)

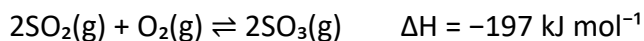
Equation:



Factor	Favours Yield?	Favours Rate?	Actual Conditions
Temperature	Low (shifts right)	High (faster collisions)	~450 °C (compromise)
Pressure	High (fewer moles RHS)	High (increased frequency)	~200 atm (high, but safe)
Catalyst	No effect on yield	Increases rate	Iron catalyst

2. Contact Process (Sulfuric acid production)

Equation:



Factor	Favours Yield?	Favours Rate?	Actual Conditions
Temperature	Low	High	~450 °C (compromise)
Pressure	High (fewer moles RHS)	High	~2 atm (sufficient yield at low cost)
Catalyst	No effect on yield	Increases rate	V ₂ O ₅ (vanadium(V) oxide)

D. Important Economic and Practical Considerations

Factor	Why It Matters
Higher pressure	Increases cost of equipment and energy
Lower temperature	Increases yield (exothermic), but slows rate
Catalysts	Increase rate, reduce energy consumption
Continuous removal of product	Shifts equilibrium towards products

The yield might be very low like 5% in order to maintain a high **rate of reaction**. This is acceptable because the **unreacted reactants** can be recycled and used back in the reactor reducing waste and improving overall efficiency. For example, in the **Haber process** any **unreacted nitrogen and hydrogen gases** are **recycled** back into the reaction vessel, so very little is wasted